

# Geometric & topological errors, at a glance

The recurring ways a shapefile or GeoJSON goes wrong — and the Python/R call that catches each one. Keep this next to your next QA pass.

## 01 GEOMETRIC VALIDITY — IS THE SHAPE ITSELF WELL-FORMED?

### ● Self-intersection

The boundary crosses itself — a "bowtie" instead of one closed area. The most common case, often from imprecise digitizing.

Spot it: `geom.is_valid` → `False`, reason mentions **"Self-intersection"**

### ● Hole outside boundary

An inner ring (e.g. a lake) extends even partly outside the outer boundary it's supposed to sit inside.

Spot it: reason mentions **"Interior is disconnected"** or **"Hole lies outside shell"**

### ● Self-touching ring

The boundary touches itself at a single point without crossing — splits the shape into two lobes joined by one vertex.

Spot it: reason mentions **"Ring Self-intersection"**

### ● Overlapping holes

Two inner rings of the same polygon intersect each other — the "no-go" area is ambiguous where they overlap.

Spot it: reason mentions **"Interior is disconnected"**

## → CHECK & FIX, SAME THREE MOVES IN BOTH LANGUAGES

### PYTHON

```
geom.is_valid          # True/False
explain_validity(geom) # why, in words
make_valid(geom)       # rebuild it valid
```

### R

```
sf::st_is_valid(x)      # TRUE/FALSE
sf::st_is_valid(x, reason = TRUE) # why, in words
sf::st_make_valid(x)    # rebuild it valid
```

Diagnosis first, repair second — `make_valid()` / `st_make_valid()` can change a shape's area or split it into multiple parts. Always check **what** changed before trusting the repaired geometry downstream.

## 02 TOPOLOGY — DO THE SHAPES RELATE TO EACH OTHER CORRECTLY?

### ● Overlap

Two polygons that should only be adjacent instead cover some of the same surface — common after a manual edit or a bad merge.

Spot it: `geom1.overlaps(geom2) → TRUE`, then measure `geom1.intersection(geom2).area`

### ● Gap

A sliver of uncovered space is left between polygons that should continuously tile an area, with no feature to claim it.

Spot it: for close, non-overlapping pairs, `union(a,b).convex_hull.difference(union(a,b))` is non-empty

### ● Sliver

A thin, geographically meaningless leftover polygon — usually the byproduct of an overlay or intersection operation.

Spot it: compactness  $4 \cdot \pi \cdot \text{area} / \text{perimeter}^2$  close to 0 (a circle scores 1)

### ● Duplicates

Two features carry the exact same geometry — inflates counts and areas silently in any aggregation downstream.

Spot it: `geom1.equals(geom2) → TRUE` (polygons or lines alike)

### ● Dangle

A line endpoint that should connect to another line (a road network, a river network) stays isolated instead.

Spot it: an endpoint with no other line endpoint within tolerance distance

## → SAME IDIOMS, R SIDE

R

```
sf::st_overlaps(a, b)           # logical matrix
sf::st_equals(a, b)             # exact-geometry duplicates
sf::st_union(sf::st_combine(x)) # dissolve, for gap-hunting
```

None of these five are covered by `is_valid` — a topologically broken dataset can be made entirely of individually-valid geometries. Validity and topology are two separate questions, checked separately.

## Try it on your own file — free, runs in your browser

Upload a GeoJSON or shapefile and get the diagnosis instantly, in both Python and R side by side. Nothing leaves your computer.

[Validator tool](#) → [Geometric validity, in depth](#) → [Topology errors, in depth](#) →